

ePGD Machine Learning — TA Session Summary

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1. Supervised vs Unsupervised Learning

A customer dataset (age, income, spending score, purchases/month, premium membership) illustrates the distinction:

Task	Type	Algorithm
Predict premium membership (Yes/No)	Supervised	Logistic Regression
Group customers into spending segments	Unsupervised	K-Means Clustering

2. Convexity, Optimization and Linear Algebra

Given $f(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2 + \theta_1\theta_2$ — check convexity and find the global minimum.

Gradient: $\nabla f = (2\theta_1 + \theta_2, \theta_1 + 2\theta_2)$. Setting $\nabla f = 0 \implies \theta_1 = \theta_2 = 0$.

Hessian $\nabla^2 f = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Leading principal minors: $2 > 0$, $\det = 3 > 0 \implies$ positive definite $\implies$ strictly convex.

Global minimum at $(\theta_1, \theta_2) = (0, 0)$ with $f(0, 0) = 0$.

3. Statistical Framework & Hypothesis Testing

Model: $y = \beta_0 + \beta_1 x + \varepsilon$, $n = 5$, $\beta_1 = 2$, $SE(\beta_1) = 0.5$. Test $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$ ($\alpha = 5\%$).

$t = \beta_1 / SE(\beta_1) = 2 / 0.5 = 4$. $df = n - 2 = 3$. Critical value $t_{0.025,3} \approx 3.182$.

$|4| > 3.182 \implies$ Reject H_0 . Feature x has a statistically significant effect on y .

4. Gradient Descent

$f(x, y) = x^2 + y^2 + xy$, $\eta = 0.2$, starting point $(1, 2)$.

$\nabla f(x, y) = (2x + y, 2y + x)$. At $(1, 2)$: $\nabla f = (4, 5)$.

Update: $x_1 = 1 - 0.2 \times 4 = 0.2$, $y_1 = 2 - 0.2 \times 5 = 1.0$.

$f(1, 2) = 7 \rightarrow f(0.2, 1) = 1.24$. Since $1.24 < 7$, objective decreases ✓

5. Convex Optimization with Equality Constraint

Minimize $x^2 + y^2$ subject to $x + y = 1$.

Lagrangian: $L(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1)$.

Stationarity: $2x + \lambda = 0$ and $2y + \lambda = 0 \implies x = y$.

Constraint $2x = 1 \implies x = y = \frac{1}{2}$. Minimum value = $\frac{1}{2}$.

6. Primal Problem — Linear Programming (Graphical Method)

Maximize $3x + 2y$ subject to $x + y \leq 4$, $x \geq 0$, $y \geq 0$.

Corner Point	Objective $3x + 2y$
$(0, 0)$	0
$(4, 0)$	12 ← Maximum
$(0, 4)$	8

Optimal: $x^* = 4$, $y^* = 0$, max value = 12.

7. KKT Conditions

Minimize x^2 subject to $x \geq 1$.

Lagrangian: $L(x, \lambda) = x^2 + \lambda(1 - x)$.

KKT Condition	Expression	Result
Stationarity	$2x - \lambda = 0$	$\lambda = 2x$
Primal feasibility	$x \geq 1$	$x^* = 1$
Dual feasibility	$\lambda \geq 0$	$\lambda^* = 2 \geq 0 \quad \checkmark$
Complementary slackness	$\lambda(1 - x) = 0$	$2(1-1) = 0 \quad \checkmark$

Optimal: $x^* = 1$, $\lambda^* = 2$, minimum value = 1.

8. Model Assessment and Selection

Three models on the same dataset — evaluated via bias-variance tradeoff:

Model	Train MSE	Test MSE	Gap	Diagnosis
A	15.2	18.7	3.5	Moderate variance
B	8.5	25.3	16.8	Overfitting (high variance)
C	12.1	13.4	1.3	Best generalisation $\checkmark$

Select Model C: lowest test MSE (13.4) and smallest gap (1.3) — best bias-variance tradeoff. Model B overfits despite low training error.

9. Bootstrapping

Bootstrap resampling draws n samples with replacement from the original dataset of size n .

Key Probability Calculation

$P(x_i \text{ not selected in one draw}) = (n-1)/n$.

Over n independent draws: $P(x_i \text{ absent}) = ((n-1)/n)^n$

For $n = 100$: $(99/100)^{100} \approx 0.366$

As $n \rightarrow \infty$: $((n-1)/n)^n \rightarrow e^{-1} \approx 0.368$

Quantity	Value
Fraction absent from each bootstrap sample	$\approx 36.8\%$
Fraction present (unique observations)	$\approx 63.2\%$
Bootstrap samples used	$B = 1000$

Implication for Model Validation

The $\approx 36.8\%$ of observations excluded from each bootstrap sample form a natural validation set — the basis of the out-of-bag (OOB) error used in Random Forests. With $B = 1000$ samples, bootstrap gives robust estimates of model performance and parameter uncertainty.

10. In-Sample Model Selection Criteria

Setup: linear regression, $k = 4$ predictors, $n = 50$ observations, training $MSE = 25$, $\sigma^2 = 30$.
 $RSS = n \times MSE = 50 \times 25 = 1250$. Parameters including intercept: $k+1 = 5$.

Mallow's C_p

$$C_p = RSS/\sigma^2 - n + 2(k+1) = 1250/30 - 50 + 10 = 41.67 - 50 + 10 = 1.67$$

$C_p \approx k+1$ indicates appropriate complexity and no significant overfitting.

AIC (Akaike Information Criterion)

$$AIC = n \cdot \ln(RSS/n) + 2(k+1) = 50 \cdot \ln(25) + 10 = 160.9 + 10 = 170.9$$

BIC (Bayesian Information Criterion)

$$BIC = n \cdot \ln(RSS/n) + \ln(n) \cdot (k+1) = 50 \cdot \ln(25) + \ln(50) \times 5 = 160.9 + 19.6 = 180.5$$

Criterion	Value	Penalty Term	Key Property
Mallow's C_p	1.67	$2(k+1)/\sigma^2$	Compare vs $k+1$; scale-dependent
AIC	170.9	$2(k+1)$	Asymptotically efficient
BIC	180.5	$\ln(n) \cdot (k+1)$	Consistent; favours simpler models

Interpretation: Lower values are preferred for all three criteria. $BIC > AIC$ because it penalises complexity more ($\ln(n)$ vs 2 per parameter). For large n , BIC tends to select sparser models than AIC.